Use Case 1: Register / Login

Goal	Register / Login
Actor	Student
Main flow	1. Student opens the Web Based ML Model Evaluator. 2. Student selects **Register** to create an account or **Login** to access an existing account. 3. Student enters the required credentials. 4. System validates the submitted information. 5. System creates the account or authenticates the student. 6. System displays the student dashboard. 7. Student can proceed to dataset upload and model evaluation.
Alternate Flow	**Alternative Flow 1A – Existing Account** 1. Student selects **Login** instead of Register. 2. Student enters valid credentials. 3. System authenticates the student and displays the dashboard.

Use Case 2: Upload Dataset

Goal	Upload Dataset
Actor	Student
Main flow	1. Student clicks **Choose File**. 2. File browser opens. 3. Student selects a valid CSV file, such as students.csv. 4. Student clicks **Upload**. 5. System validates the CSV format, file size, and structure. 6. System stores the dataset and calculates metadata such as rows and columns. 7. System displays upload confirmation with the dataset metadata. 8. Student proceeds to Preview Dataset.
Alternate Flow	**Alternative Flow 2A – Replace Active Dataset** 1. System detects an active dataset and asks whether to replace it or preview it. 2. Student chooses to replace the dataset and selects a new CSV file. 3. Flow continues from Main Flow Step 3.

Use Case 3: Preview Dataset

Goal	Preview Dataset
Actor	Student
Main flow	1. System retrieves the uploaded dataset. 2. System displays the first few rows in a table. 3. System displays column names and available dataset information. 4. Student reviews the sample data to confirm that the dataset was read correctly. 5. Student proceeds to Preprocess Data.
Alternate Flow	**Alternative Flow 3A – Return to Upload** 1. Student decides that the uploaded dataset is not suitable. 2. Student returns to the upload page. 3. Student selects another valid CSV file. 4. Flow continues from Upload Dataset.

Use Case 4: Preprocess Data

Goal	Preprocess Data
Actor	Student
Main flow	1. System displays preprocessing options for missing values, categorical encoding, and feature scaling. 2. Student reviews the dataset and selects the required preprocessing options. 3. Student clicks **Apply Preprocessing**. 4. System validates the selected options. 5. System applies missing-value handling, categorical encoding, and feature scaling. 6. System displays preprocessing completion confirmation. 7. System stores or caches the processed dataset for subsequent steps. 8. Student proceeds to Select Target Column.
Alternate Flow	**Alternative Flow 4A – Rerun Preprocessing** 1. Student returns to preprocessing to change an option. 2. Student modifies the preprocessing settings. 3. Student clicks **Apply Preprocessing** again. 4. Flow continues with the updated processed dataset without requiring another upload.

Use Case 5: Select Target Column

Goal	Select Target Column
Actor	Student
Main flow	1. System displays the available columns from the preprocessed dataset. 2. System shows column names, data types, and sample values. 3. Student selects the target column, such as **Result** for classification or **Price** for regression. 4. Student clicks **Set as Target**. 5. System validates the selected column. 6. System designates the selected column as target **y** and the remaining suitable columns as features **X**. 7. System detects whether the problem is classification or regression. 8. System displays target-selection confirmation. 9. Student proceeds to Set Train-Test Split.
Alternate Flow	**Alternative Flow 5A – Change Target** 1. Student returns to target selection. 2. Student selects a different valid column. 3. System discards the previous target selection. 4. The newly selected column becomes the active target.

Use Case 6: Set Train-Test Split

Goal	Set Train-Test Split
Actor	Student
Main flow	1. System displays a train-test split slider or input field. 2. System shows the default **80% training / 20% testing** ratio. 3. Student reviews or adjusts the ratio, such as 70-30. 4. Student clicks **Apply Split**. 5. System validates the split configuration. 6. System partitions the dataset into training and testing data while keeping features and target values aligned. 7. System displays the number of training and testing rows. 8. Student proceeds to Select ML Algorithm.
Alternate Flow	**Alternative Flow 6A – Adjust Split and Retrain** 1. Student returns to split configuration after viewing results. 2. Student changes the split ratio. 3. Student clicks **Apply Split**. 4. Subsequent algorithm selection and training use the new split without requiring another upload.

Use Case 7: Select ML Algorithm

Goal	Select ML Algorithm
Actor	Student
Main flow	1. System displays the supported machine learning algorithms. 2. Student reviews the available algorithms for the detected problem type. 3. Student selects an algorithm, such as Logistic Regression, Decision Tree, Random Forest, or KNN. 4. System records the selected algorithm. 5. System displays the selected model configuration. 6. Student proceeds to Train Model.
Alternate Flow	**Alternative Flow 7A – Select Another Algorithm** 1. Student changes the selected algorithm before training. 2. System updates the model configuration. 3. Student continues to Train Model with the newly selected algorithm.

Use Case 8: Train Model

Goal	Train Model
Actor	Student and ML Service
Main flow	1. Student reviews the dataset, target, split, and selected algorithm. 2. Student clicks **Train Model**. 3. System sends the training request and prepared training data to the **ML Service**. 4. ML Service initializes the selected machine learning algorithm. 5. ML Service trains the model using the training partition. 6. ML Service generates predictions using the test partition and calculates the required evaluation metrics. 7. System stores the trained model and associated results. 8. System displays a training-completion message to the student. 9. Student proceeds to Evaluate Model or Compare Models.
Alternate Flow	**Alternative Flow 8A – Retrain** 1. Student changes the algorithm or train-test split after a completed run. 2. Student starts training again. 3. ML Service trains the model using the updated configuration. 4. New results are stored for comparison.

Use Case 9: Evaluate Model

Goal	Evaluate Model
Actor	Student and Instructor
Main flow	1. System retrieves the evaluation results for a trained model. 2. System displays the appropriate metrics. 3. For classification, the system displays Accuracy, Precision, Recall, F1-score, and Confusion Matrix. 4. For regression, the system displays RMSE, MAE, and R^2 Score. 5. System generates relevant visualizations such as metric charts, confusion matrix, or residual plots. 6. Student reviews the evaluation results. 7. Instructor can view the available evaluation results for review.
Alternate Flow	**Alternative Flow 9A – View Results Again** 1. Student or Instructor opens a previously generated evaluation result. 2. System retrieves the stored metrics and visualizations. 3. The results are displayed without retraining the model.

Use Case 10: Compare Models

Goal	Compare Models
Actor	Student
Main flow	1. System displays trained models available for the current dataset. 2. Student selects at least two trained models. 3. Student clicks **Compare**. 4. System retrieves the evaluation results for the selected models. 5. System displays the model performances side by side using a metric table and comparison chart. 6. Student reviews the results and identifies the better-performing model. 7. Student can select a model for download.
Alternate Flow	**Alternative Flow 10A – Compare More Than Two Models** 1. Student selects three or more compatible trained models. 2. System displays the selected models together in the comparison view. 3. Student reviews the comparative metrics and selects a preferred model.

Use Case 11: Download Model / Report

Goal	Download Model / Report
Actor	Student
Main flow	1. System displays download options for the trained model and evaluation report. 2. Student selects **Download Trained Model** or **Download Evaluation Report**. 3. System retrieves the requested model artifact or generates the evaluation report. 4. System sends the file to the student's browser. 5. Student saves the downloaded file locally. 6. The report contains model information, dataset information, preprocessing, split ratio, evaluation metrics, and relevant visualizations.
Alternate Flow	**Alternative Flow 11A – Report Only** 1. Student selects **Download Evaluation Report**. 2. System generates the PDF report from the stored evaluation results. 3. System sends the PDF to the student's browser. 4. Student saves the report locally.

Use Case 12: Manage Users

Goal	Manage Users
Actor	Admin
Main flow	1. Admin logs in to the administration area. 2. System displays the user-management interface. 3. Admin views the registered users. 4. Admin selects a user account to manage. 5. Admin performs an allowed management action, such as viewing, updating, or deactivating the account. 6. System saves the updated user information. 7. System displays confirmation of the completed action.
Alternate Flow	**Alternative Flow 12A – Review User Information** 1. Admin selects a user from the user list. 2. System displays the user's available account information and associated activity metadata. 3. Admin returns to the user list.

Use Case 13: Manage Datasets

Goal	Manage Datasets
Actor	Admin
Main flow	1. Admin opens the dataset-management area. 2. System displays stored dataset records and metadata. 3. Admin selects a dataset record. 4. Admin reviews or manages the dataset record according to administrative permissions. 5. System applies the requested management action. 6. System displays confirmation.
Alternate Flow	**Alternative Flow 13A – Review Dataset Metadata** 1. Admin selects a dataset record. 2. System displays dataset name, ownership information, and available metadata. 3. Admin returns to the dataset list.

Use Case 14: View System Logs

Goal	View System Logs
Actor	Admin
Main flow	1. Admin opens the system-log interface. 2. System retrieves available application activity logs. 3. System displays relevant events such as uploads, model training activity, and administrative actions. 4. Admin reviews the logs to monitor system activity. 5. Admin returns to the administration dashboard.
Alternate Flow	**Alternative Flow 14A – Review Specific Activity** 1. Admin selects or filters a relevant log entry. 2. System displays the associated event information. 3. Admin continues monitoring system activity.